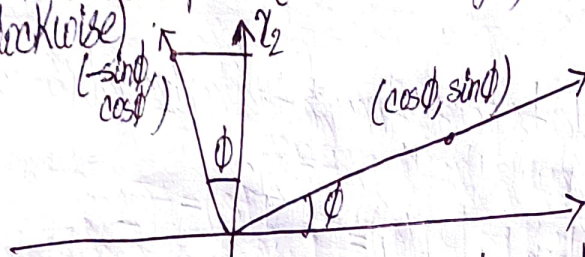


3. $T: \mathbb{R}^2 \rightarrow \mathbb{R}^2$ rotates points (about the origin) through $3\pi/2$ radians (counterclockwise)

Solution:



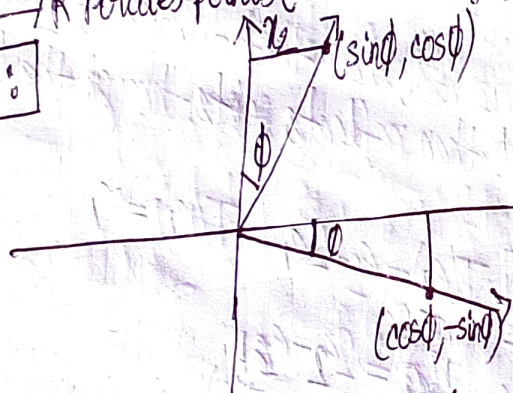
The transformation matrix is:

$$\begin{bmatrix} \cos \phi & -\sin \phi \\ \sin \phi & \cos \phi \end{bmatrix}$$

$$= \begin{bmatrix} \cos \frac{3\pi}{2} & -\sin \frac{3\pi}{2} \\ \sin \frac{3\pi}{2} & \cos \frac{3\pi}{2} \end{bmatrix} = \begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix}$$

4. $T: \mathbb{R}^2 \rightarrow \mathbb{R}^2$ rotates points (about the origin) through $-\pi/4$ radians (clockwise)

Solution:



By Theorem 10, the transformation matrix is:

$$\begin{bmatrix} \cos \phi & \sin \phi \\ -\sin \phi & \cos \phi \end{bmatrix}$$

$$= \begin{bmatrix} \cos \pi/4 & \sin \pi/4 \\ -\sin \pi/4 & \cos \pi/4 \end{bmatrix} = \begin{bmatrix} 1/\sqrt{2} & 1/\sqrt{2} \\ -1/\sqrt{2} & 1/\sqrt{2} \end{bmatrix}$$

5. $T: \mathbb{R}^2 \rightarrow \mathbb{R}^2$ is a vertical shear transformation that maps e_1 into $e_1 - 2e_2$ but leaves the vector e_2 unchanged.

Solution:

$$T(e_1) = e_1 - 2e_2 = \begin{bmatrix} 1 \\ 0 \end{bmatrix} - 2 \begin{bmatrix} 0 \\ 1 \end{bmatrix} = \begin{bmatrix} 1 \\ -2 \end{bmatrix}$$

$$T(e_2) = e_2$$

$\therefore$ The standard matrix is: $\begin{bmatrix} 1 & 0 \\ -2 & 1 \end{bmatrix}$ (By Theorem 10)

7. $T: \mathbb{R}^2 \rightarrow \mathbb{R}^2$ first rotates points through $-3\pi/4$ radian (clockwise) and then reflects points through the horizontal x -axis.

Solution:

Transformation 1: By Theorem 10, the standard matrix is

$$\begin{bmatrix} \cos 3\pi/4 & \sin 3\pi/4 \\ -\sin 3\pi/4 & \cos 3\pi/4 \end{bmatrix}$$

$$T(e_1) = \begin{bmatrix} \cos 3\pi/4 \\ -\sin 3\pi/4 \end{bmatrix} \quad T(e_2) = \begin{bmatrix} -\sin 3\pi/4 \\ -\cos 3\pi/4 \end{bmatrix}$$

Transformation 2:

By Theorem 10, The standard matrix is:

$$\begin{bmatrix} \cos 3\pi/4 & -\sin 3\pi/4 \\ -\sin 3\pi/4 & -\cos 3\pi/4 \end{bmatrix}$$